

Anti-neurofascin antibody in patients with combined central and peripheral demyelination

Nobutoshi Kawamura,
MD, PhD*
Ryo Yamasaki, MD,
PhD*
Tomomi Yonekawa, MD
Takuya Matsushita, MD,
PhD
Susumu Kusunoki, MD,
PhD
Shigemi Nagayama, MD,
PhD
Yasuo Fukuda, MD, PhD
Hidenori Ogata, MD
Dai Matsuse, MD, PhD
Hiroyuki Murai, MD,
PhD
Jun-ichi Kira, MD, PhD

Correspondence to
Dr. Kira:
kira@neuro.med.kyushu-u.ac.jp

ABSTRACT

Objectives: We aimed to identify the target antigens for combined central and peripheral demyelination (CCPD).

Methods: We screened target antigens by immunohistochemistry and immunoblotting using peripheral nerve tissues to identify target antigens recognized by serum antibodies from selected CCPD and chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) cases. We then measured the level of antibody to the relevant antigen in 7 patients with CCPD, 16 patients with CIDP, 20 patients with multiple sclerosis, 20 patients with Guillain-Barré syndrome, 21 patients with other neuropathies, and 23 healthy controls (HC) by ELISA and cell-based assays using HEK293 cells.

Results: At the initial screening, sera from 2 patients with CCPD showed cross-like binding to sciatic nerve sections at fixed intervals, with nearly perfect colocalization with neurofascin immunostaining at the node and paranode. ELISA with recombinant neurofascin revealed significantly higher mean optical density values in the CCPD group than in other disease groups and HC. Anti-neurofascin antibody positivity rates were 86% in patients with CCPD, 10% in patients with multiple sclerosis, 25% in patients with CIDP, 15% in patients with Guillain-Barré syndrome, and 0% in patients with other neuropathies and HC. The cell-based assay detected serum anti-neurofascin antibody in 5 of 7 patients with CCPD; all others were negative. CSF samples examined from 2 patients with CCPD were both positive. In anti-neurofascin antibody-positive CCPD patients, including those with a limited response to corticosteroids, IV immunoglobulin or plasma exchange alleviated the symptoms.

Conclusion: Anti-neurofascin antibody is frequently present in patients with CCPD. Recognition of this antibody may be important, because patients with CCPD who are antibody positive respond well to IV immunoglobulin or plasma exchange. *Neurology*® 2013;81:714-722

GLOSSARY

AIDP = acute inflammatory demyelinating polyradiculoneuropathy; **AMAN** = acute motor axonal neuropathy; **Caspr** = contactin-associated protein 1; **CCPD** = combined central and peripheral demyelination; **CIDP** = chronic inflammatory demyelinating polyradiculoneuropathy; **GBS** = Guillain-Barré syndrome; **GFP** = green fluorescent protein; **HC** = healthy controls; **IgG** = immunoglobulin G; **MS** = multiple sclerosis; **NF** = neurofascin; **OCB** = oligoclonal immunoglobulin G bands; **OD** = optical density; **ON** = other neuropathies; **PNS** = peripheral nervous system; **ROC** = receiver operating characteristic.

Multiple sclerosis (MS) affects the CNS, with no involvement of the peripheral nervous system (PNS). In contrast, chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) is typically restricted to the PNS and can be considered a peripheral nerve analog of MS. However, the phenotype of CIDP is wide and the disease can be associated with CNS demyelination. Patients presenting with combined central and peripheral demyelination (CCPD) have occasionally been reported using various diagnostic names including chronic demyelinating peripheral neuropathy associated with multifocal CNS demyelination, peripheral neuropathy with MS, CIDP with CNS involvement, and relapsing demyelinating disease affecting both the CNS and PNS.¹ CCPD currently has no clear clinical definition

Supplemental data at
www.neurology.org

*These authors contributed equally to this manuscript.

From the Departments of Neurology (N.K., T.Y., T.M., H.O., D.M., H.M., J.K.) and Neurological Therapeutics (R.Y.), Neurological Institute, Graduate School of Medical Sciences, Kyushu University, Fukuoka; Department of Neurology (S.K.), School of Medicine, Kinki University, Osaka; Department of Neurology (S.N.), Kanazawa Medical University; and Department of Neurology (Y.F.), Sasebo City General Hospital, Japan.

Go to Neurology.org for full disclosures. Funding information and disclosures deemed relevant by the authors, if any, are provided at the end of the article.

and appears to encompass heterogeneous conditions including acute, relapsing, and chronic subtypes.

This combined demyelinating condition has raised the issue of whether this is a unique condition due to a common immunopathogenic mechanism, or simple coincidence of restricted demyelinating disorders (MS and CIDP). Several clinical, laboratory, and neuroimaging features in patients with CCPD are atypical for MS, including bilateral optic neuritis, an absence of oligoclonal immunoglobulin G (IgG) bands (OCB), and gray matter involvement.^{2–4} Therefore, a distinct immune-mediated mechanism causing both central and peripheral demyelination is postulated in this entity; however, no antigenic target common to the CNS and PNS has yet been identified. In this study, as one of target antigens for CCPD, we identified neurofascin, which is a member of the L1 subgroup of adhesion molecules expressed at the nodes of Ranvier and the paranodes in both the CNS and PNS,⁵ and characterized the clinical profiles of anti-neurofascin antibody–positive CCPD patients.

METHODS Patients. We obtained clinical data and sera from 7 patients with CCPD, 16 patients with CIDP based on the research criteria for diagnosis of CIDP,⁶ 20 patients with MS according to the revised McDonald criteria,⁷ 20 patients with Guillain-Barré syndrome (GBS) (10 with acute motor axonal neuropathy [AMAN],⁸ 7 with acute inflammatory demyelinating polyneuropathy [AIDP],⁸ and 3 unclassified), 21 patients with other neuropathies (ON), and 23 healthy controls (HC). Written informed consent was obtained. All patients with CCPD were confirmed to have CNS lesions suggestive of demyelination by MRI and demyelinating neuropathy by peripheral nerve conduction studies. Among them, 3 cases (patients 1, 2, and 6 in the table) were diagnosed with CCPD at Kyushu University Hospital between 2001 and 2010 and used for initial screening. The other 4 cases were consecutively referred to Kyushu University from the coauthors' institutions in 2011 for the anti-neurofascin antibody assay. All CCPD samples were subjected to immunoblotting, ELISA, and cell-based assays. Only one patient in the MS group had abnormal nerve conduction study findings due to carpal tunnel syndrome. Fourteen patients in the CIDP group showed typical symmetric distal and proximal weakness, while 2 patients had asymmetric involvement. Four patients with CIDP had abnormalities on brain MRI: 3 with small ischemic lesions and one with mild hydrocephalus. ON included inherited, vasculitic, and diabetic neuropathies. Samples were collected from patients receiving no immunologic treatments. There were no significant differences in the age and sex of each group except for patients with CCPD, who were younger than those in the other groups.

Immunofluorescence assay. The sciatic nerves of female Sprague-Dawley rats were collected and frozen immediately after dissection. Longitudinal sections of the sciatic nerves were fixed

for 3 minutes in 10% formalin just before immunohistochemical staining. We performed an indirect immunofluorescence assay on these samples using sera from human subjects (diluted 1:60) followed by detection with a fluorescein-conjugated goat antibody to human IgG (Southern Biotechnology Associates, Birmingham, AL). We also undertook double immunostaining to characterize the localization of patients' IgG binding sites, using rabbit IgG antibodies specific for neurofascin, gliomedin (Abcam Plc., Cambridge, UK), or contactin-associated protein 1 (Caspr) (Millipore Corporation, Temecula, CA), together with an Alexa 594–conjugated antibody to rabbit IgG as a detector reagent (Southern Biotechnology Associates). Images were captured using a fluorescence microscope (Biozero, BZ-8000; Keyence, Osaka, Japan).

Extraction of tissue proteins. Detailed methods for the extraction of tissue proteins are described in the e-Methods on the *Neurology*[®] Web site at www.neurology.org.

Immunoblotting and immunoabsorption. Detailed methods for immunoblotting using extracted PNS and CNS proteins and recombinant rat neurofascin protein and immunoabsorption by recombinant neurofascin protein are described in the e-Methods.

Cell-based assay. We also performed a cell-based anti-neurofascin antibody detection assay using neurofascin-transfected cells. Human embryonic kidney 293 cells maintained in medium containing 10% fetal calf serum were seeded at 5,000 cells/well onto 8-well chamber slides 24 hours before transfection. Because neurofascin has 2 major forms, neurofascin 186 (NF186) and neurofascin 155 (NF155), the cells were transfected with 100 ng/well of either green fluorescent protein (GFP)-NF155 or GFP-NF186 fusion protein expression vector containing a full-length cDNA encoding human NF155 or NF186, respectively (OriGene Technologies, Inc., Rockville, MD), using FuGENE6 transfection reagent (Roche, Basel, Switzerland). After blocking with 10% goat serum, the GFP-NF155–expressing cells were incubated with human serum (1:100) or CSF samples (1:1) diluted with Dulbecco's modified Eagle medium for 1 hour at 37°C without cell fixation, washed in phosphate-buffered saline, and then visualized using an Alexa 594–conjugated goat antibody to human IgG (Invitrogen, Carlsbad, CA). The fluorescence of unfixed cells was observed using a confocal laser scanning microscope (Nikon A1). The anti-neurofascin antibody assay was performed at least twice for each sample, and those that gave a positive result twice were deemed to be positive.

ELISA. We coated polystyrene microtiter ELISA plates (ICN Biomedicals, Inc., Costa Mesa, CA) with recombinant rat neurofascin protein (0.2 µg/well). After blocking, 50 µL of serum (1:200) was applied to the wells at room temperature and plates were incubated for 1 hour. After washing, we applied a horseradish peroxidase–conjugated goat antibody to human IgG (1:1000; Southern Biotechnology Associates) and developed the plates using *o*-phenylenediamine dihydrochloride (Sigma, St. Louis, MO). The optical density (OD) was quantified at 450 nm using an Immunomini NJ-2300 plate reader (Inter Med, Osaka, Japan).

Statistical analysis. Data are expressed as means and SEM. We used 1-way analysis of variance for comparisons of ages, and Fisher exact probability test for comparisons of the frequencies of sex in each group. We also used 1-way analysis of variance followed by Dunnett multiple comparison tests for comparisons of the mean OD values in each group (figure 3B). Receiver operating characteristic (ROC) curves were generated and area under the curve was calculated to estimate cutoff OD values for the anti-neurofascin ELISA assay to accurately discriminate patients with CCPD. Statistical analyses were performed using JMP 9.0.2, and the threshold for significance was set at $p < 0.05$.

Table Presentations of patients with CCPD

	Patient 1	Patient 2	Patient 3	Patient 4	Patient 5	Patient 6	Patient 7
Age, y/sex	16/F	25/M	57/F	22/M	44/M	23/F	38/F
Initial symptoms	CNS	PNS and CNS	PNS	PNS	CNS and PNS	CNS	CNS and PNS
Age at CNS onset, y	16	23	56	18	44	10	38
Age at PNS onset, y	16	23	48	10	44	23	38
Temporal gap between CNS and PNS onsets	Sequential (3 mo apart)	Simultaneous	Long interval (8 y)	Long interval (8 y)	Simultaneous	Long interval (10 y)	Simultaneous
Laboratory data							
Anti-NF155 antibody (cell-based assay) (serum)	+	+	+	+	+	–	–
Anti-NF155 antibody (ELISA)	+	+	+	+	+	–	+
CSF protein, mg/dL	205	299	55	97	86	83	89
OCB	–	–	+	–	–	–	–
MRIs							
Brain	Multifocal WML; Gd-enhanced	Diffuse WML; No Gd enhancement	Multifocal WML; Gd-enhanced	Multifocal WML; No Gd enhancement	Multifocal WML	Multifocal WML; Gd-enhanced	Multifocal WML
Spinal cord	Normal	Normal	Multifocal short lesions	Multifocal short lesions	Normal	Multifocal lesions	Normal
Cauda equina	Gd-enhanced; Hypertrophy	Normal	ND	ND	Normal	Normal	Normal
McDonald criteria for MS	Fulfilled	Unfulfilled	Fulfilled	Fulfilled	Fulfilled	Fulfilled	Fulfilled
NCS							
Prolonged DL	+	+	+	+	+	–	+
MCV slowing	+	+	+	+	+	+	+
CB or TD	+	+	+	+	–	+	+
Prolonged F-wave latency	+	+	ND	ND	+	+	+
EFNS/PNS criteria for CIDP	Definite	Definite	Definite	Definite	Definite	Definite	Definite
VEPs	Normal	Bilaterally prolonged	Unilaterally prolonged	Unilaterally prolonged	ND	Unilaterally prolonged	ND
Treatment							
Corticosteroids	Effective for CNS but ineffective for PNS lesions	Ineffective	Partially effective	Effective for CNS lesions	Partially effective	Effective for both CNS and PNS lesions	Effective for both CNS and PNS lesions
IVIg	Effective	Effective	Effective	ND	Effective	ND	ND
PE	Effective	Effective	Ineffective	ND	ND	ND	ND
Other drugs	–	–	AZT, INFβ-1a/b (insufficient)	INFβ-1b	–	INFβ-1a	–

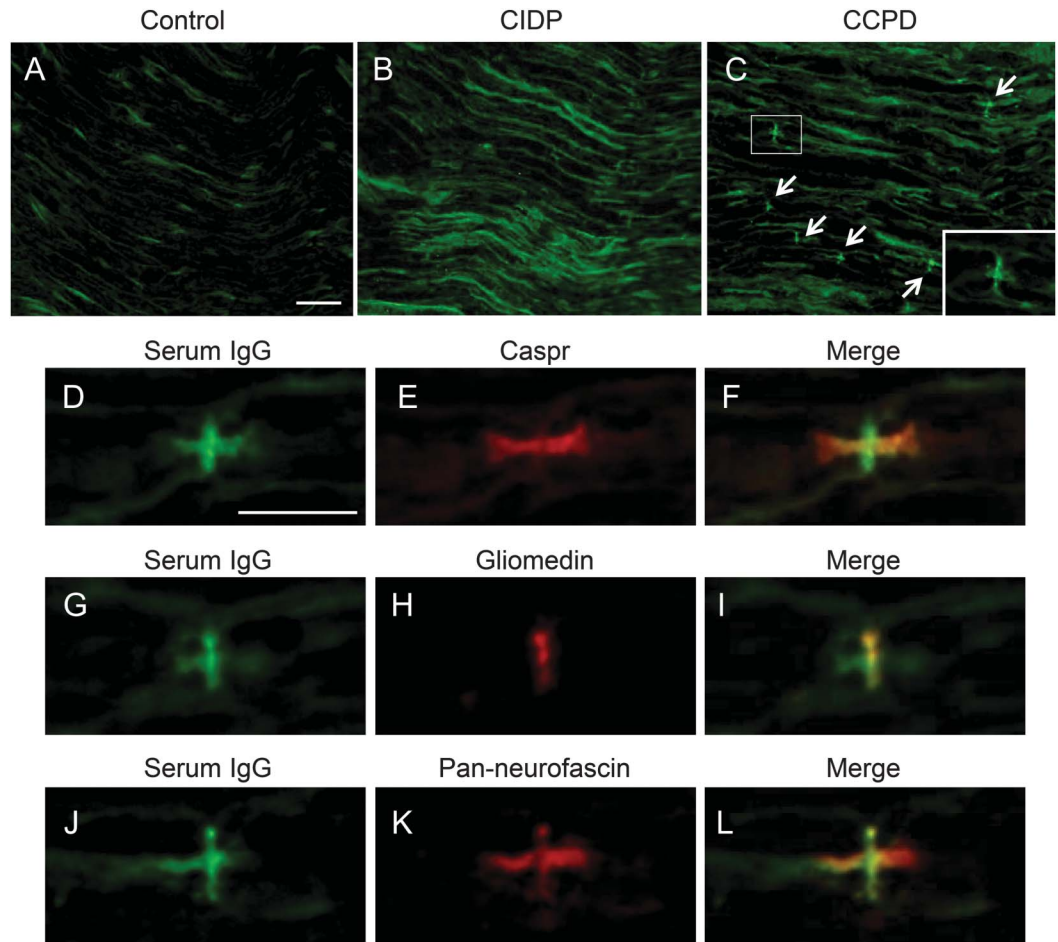
Abbreviations: AZT = azidothymidine; CB = conduction block; CCPD = combined central and peripheral demyelination; CIDP = chronic inflammatory demyelinating polyradiculoneuropathy; DL = distal latency; EFNS/PNS = European Federation of Neurological Societies and the Peripheral Nerve Society; Gd = gadolinium; INFβ = interferon β; IVIg = IV immunoglobulin; MCV = motor conduction velocity; MS = multiple sclerosis; NCS = nerve conduction study; ND = not done; NF = neurofascin; OCB = oligoclonal immunoglobulin G bands; PE = plasma exchange; PNS = peripheral nervous system; TD = temporal dispersion; VEP = visual evoked potential; WML = white matter lesion.

Standard protocol approvals. The research protocol for this retrospective study and the data privacy procedures for consented samples were approved by the Kyushu University ethics committee (no. 24-100).

RESULTS Screening and identification of target proteins in patients with CCPD and CIDP. During screening by immunofluorescence assay, sera from controls showed

no significant immunoreactivity toward the rat sciatic nerve (figure 1A). Four patients with CIDP showed diffuse binding of serum IgG to the myelin sheath (figure 1B). Two patients with CCPD (patients 1 and 2) showed a characteristic pattern of serum IgG binding to sections of sciatic nerves, with a cross-like appearance at fixed intervals (figure 1C).

Figure 1 Screening of target proteins in patients with CCPD



(A–C) Immunofluorescence analysis of rat sciatic nerve reacted with patients' sera. Note the cross-like appearance in panel C (arrows) and inset. (A) Control. (B) Patient with CIDP. (C) Patient with CCPD (patient 2 in the table). Bar: 100 μ m. (D–L) Double immunofluorescence analysis of the nodes of Ranvier in rat sciatic nerves reacted with CCPD patient's serum and each specific antibody. (D, G, J) Serum from patient with CCPD. (E, H, K) Specific antibody indicated. (F, I, L) Merged images. Almost perfect colocalizations are seen in panels J–L. Bar: 10 μ m. CCPD = combined central and peripheral demyelination; CIDP = chronic inflammatory demyelinating polyradiculoneuropathy; IgG = immunoglobulin G.

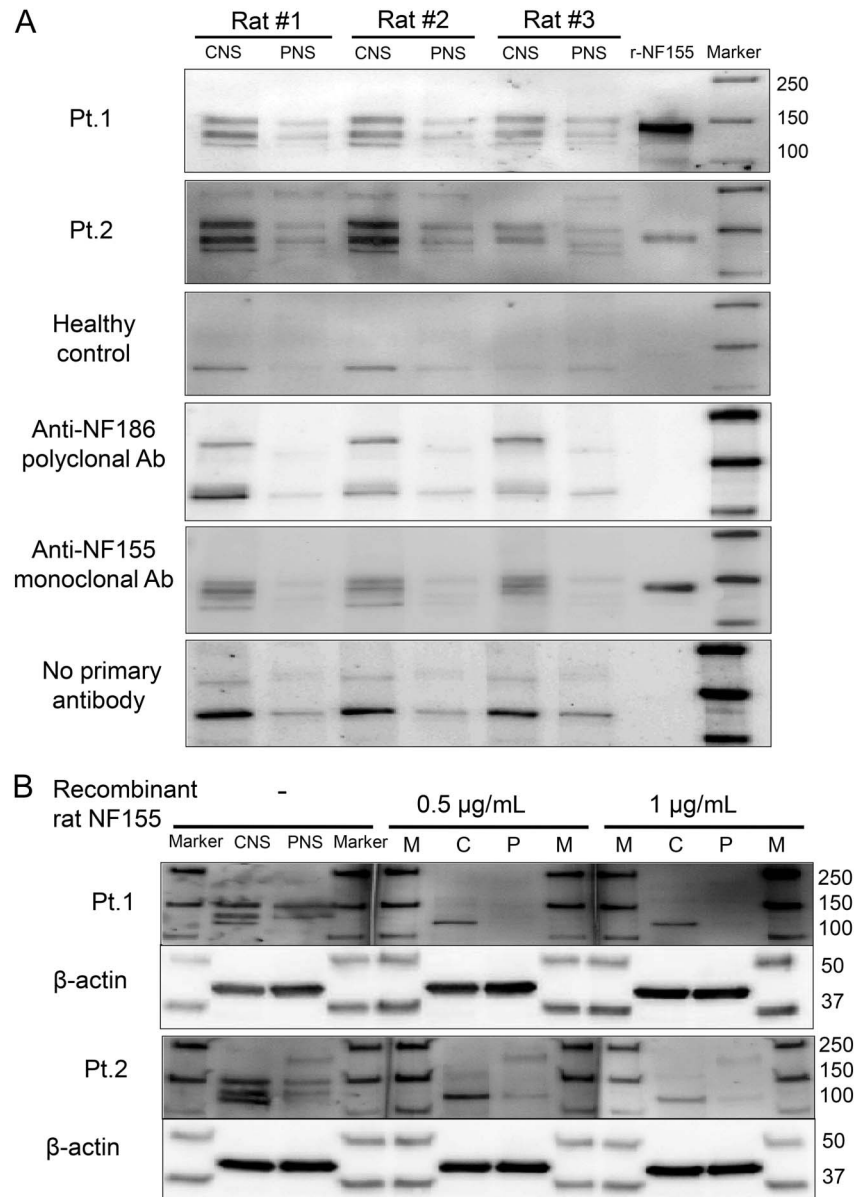
Dual labeling experiments for serum IgG binding and nodal or paranodal proteins revealed a partial overlay of serum IgG binding with gliomedin labeling at the node of Ranvier and with Caspr labeling at the paranode. By contrast, pan-neurofascin immunostaining showed nearly perfect colocalization with serum IgG binding at both the node and paranode (figure 1, D–L).

Immunoblotting experiments revealed that the serum IgG from CCPD patient 1 bound to both rat sciatic nerve (PNS) and spinal cord (CNS) lysates as 2 dense bands and one additional thin band between 100 and 150 kD (figure 2A). Serum from patient 2 also showed 2 thick bands and one thin band between 100 and 150 kD, and an additional faint band between 150 and 250 kD. The mobility of the 2 dense bands stained by the patients' sera was in accord with that stained by anti-NF155 monoclonal antibody, but not with that stained by anti-NF186 polyclonal antibody. The 2 dense bands disappeared

when the patients' sera were preincubated with recombinant NF155 while the lower-molecular-weight thin band in patients 1 and 2 and the higher-molecular-weight faint band in patient 2 still existed (figure 2B). NF155 comprises 2 isoforms, termed neurofascin 155 high and low (NF155H and NF155L, respectively), which differ in terms of their N-linked glycosylation.⁹ These findings suggest that the 2 dense bands correspond to NF155H and NF155L, whereas the lower-molecular-weight band in patients 1 and 2 and the higher-molecular-weight band in patient 2 were non-specific. The other CCPD cases showed no clear bands on immunoblotting.

Anti-neurofascin antibody in the cell-based assay and ELISA in patients with CCPD and other neurologic disorders and HC. We examined the presence/absence of IgG antibody to neurofascin in patients with CIDP, MS, AMAN, AIDP, and ON and HC using

Figure 2 Identification of anti-neurofascin antibody in sera of patients with CCPD

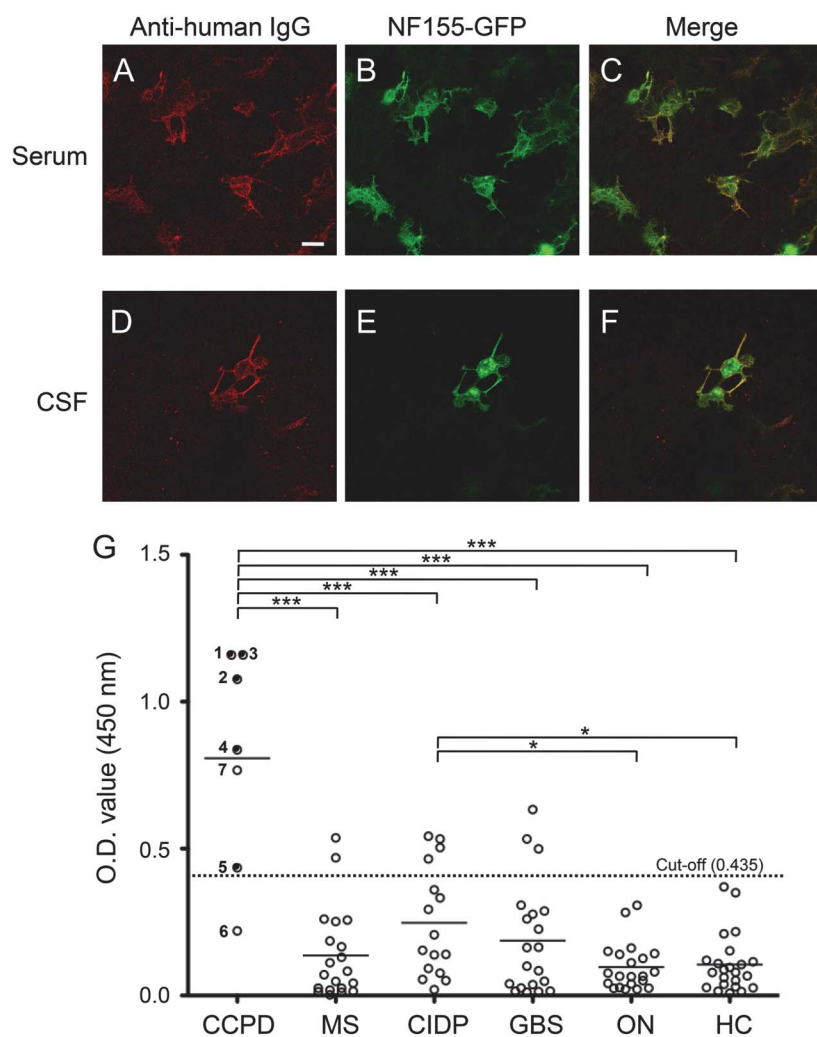


(A) Immunoblotting analysis of sera of patients with CCPD, serum from healthy control, anti-NF186 polyclonal antibody, and anti-NF155 monoclonal antibody using rat CNS and PNS tissues ($n = 3$) and rat recombinant neurofascin protein (r-NF155). Pt.1 and 2 correspond to patients 1 and 2 in the table, respectively. Three bands are visible between 100 and 150 kD in patients 1 and 2. The mobility of the 2 dense bands stained by the patients' sera is in accord with that stained by anti-NF155 monoclonal antibody, but not with that stained by anti-NF186 polyclonal antibody. The bands of the recombinant NF155 protein and rat brain NF155 do not line up perfectly because recombinant NF155 is His-tagged, which causes it to have a distinct electrophoretic mobility. The NF186 band is visible only in rat CNS samples because NF186 is mainly expressed in CNS tissues.¹⁸ (B) Immunoabsorption experiments using sera from patients 1 and 2. By preincubation of diluted patients' sera (1:100) with recombinant rat NF155 at 2 different concentrations, the 2 upper dense bands are not visible while the lower thin band in patients 1 and 2 and the higher-molecular-weight thin band between 150 and 250 kD in patient 2 still exist, suggesting that the patients' antibodies specifically bind to NF155 while the lower thin band in both patients and the higher-molecular-weight thin band in patient 2 are nonspecific. Ab = antibody; CCPD = combined central and peripheral demyelination; NF = neurofascin; PNS = peripheral nervous system.

a cell-based assay and ELISA. In the cell-based assay, antibodies against NF155 were detected in sera from 5 patients with CCPD (patients 1–5 in the table), whereas all others were negative (figure 3, A–C). In patients 1 and 2, further analyses of CSF samples

confirmed both to be positive for anti-NF155 antibody (figure 3, D–F). However, these patients' sera were negative for NF186 in the cell-based assay (data not shown). In the ELISA for NF155, among the 5 patients with CCPD who were positive for anti-neurofascin

Figure 3 Detection of anti-neurofascin antibody in patients with CCPD



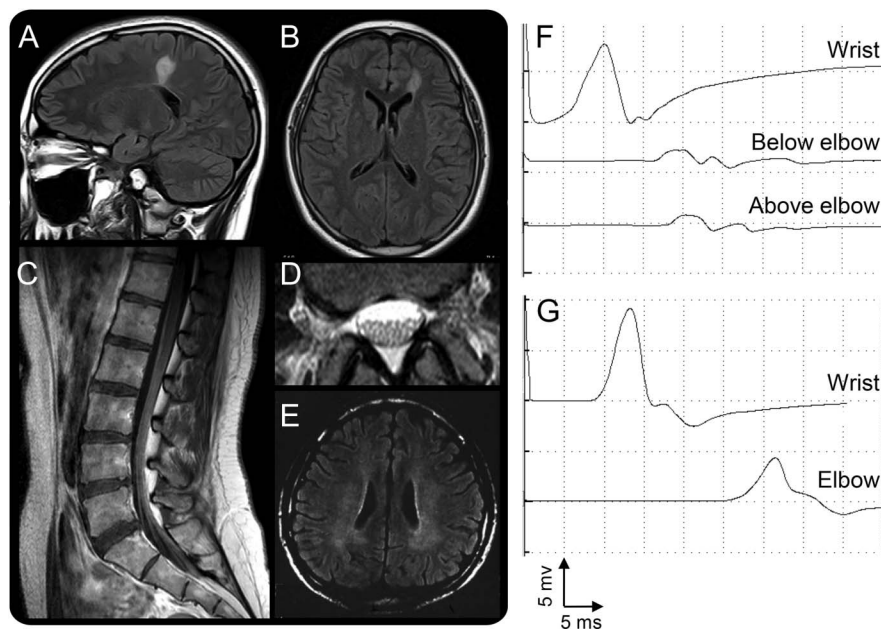
(A-F) Cell-based assay for anti-neurofascin antibody in CCPD patient's serum and CSF. Serum (1:100 dilution) (A-C) and CSF (1:1) (D-F) from patient 2 both react well to HEK293 cells expressing human NF155 protein with a GFP-tag on the cell surface. Bar: 10 μ m. (G) Anti-neurofascin antibody titer determined by ELISA. Filled dots: patients who were also positive by the cell-based assay (figure 2B). Empty dots: patients negative by the cell-based assay. Mean values are indicated in each column. The number next to each dot corresponds to the patient number in the table. The mean OD values and standard errors for each group were as follows: CCPD: 0.81 ± 0.14 (n = 7); MS: 0.14 ± 0.03 (n = 20); CIDP: 0.25 ± 0.05 (n = 16); GBS: 0.19 ± 0.04 (n = 20); ON: 0.10 ± 0.02 (n = 21); and HC: 0.11 ± 0.02 (n = 23). * $p < 0.05$, *** $p < 0.001$. CCPD = combined central and peripheral demyelination; CIDP = chronic inflammatory demyelinating polyradiculoneuropathy; GBS = Guillain-Barré syndrome; GFP = green fluorescent protein; HC = healthy controls; IgG = immunoglobulin G; MS = multiple sclerosis; NF = neurofascin; OD = optical density; ON = other neuropathies.

antibody in the cell-based assay, 3 showed very high serum OD values (figure 3G, patients 1–3 in the table) and 2 had medium OD values (patients 4 and 5). One patient with CCPD (patient 7) who showed a negative result on the cell-based assay had a medium OD value by ELISA. Mean OD values in patients with CCPD were significantly higher than those in other disease groups and HC (figure 3G). Logistic regression analysis showed that when OD values of samples from patients with CCPD were compared with samples from patients

with other neurologic disorders, including MS, CIDP, GBS, and ON, and samples from HC, the area under the ROC curve was 0.95. When the cutoff point was estimated from the ROC curve, the ideal cutoff OD value was estimated as 0.435. According to the cutoff point for the OD value, the numbers of anti-neurofascin antibody–positive patients in each disease category were as follows: 6/7 patients with CCPD (86%); 2/20 patients with MS (10%); 4/16 patients with CIDP (25%); 2/10 patients with AMAN (20%); 1/7 patients with AIDP (14.3%); 0/21 patients with ON (0%); and 0/23 HC (0%). Sera from all CIDP, MS, AMAN, and AIDP patients who had OD values higher than the cutoff level by ELISA were found to be negative by the cell-based assay (figure 3G, empty dots).

Case presentations and comparison of clinical features between anti-neurofascin–positive and –negative CCPD patients. The clinical findings of the 7 patients with CCPD are summarized in the table, and each case is described in appendix e-1. The age at onset in the 5 patients with CCPD who were anti-neurofascin antibody positive by the cell-based assay ranged from 16 to 48 years, and both sexes were involved (male:female, 2:3). The CNS and PNS were involved simultaneously or with a short interval in 3, whereas 2 showed a long interval between CNS and PNS involvement. All had high CSF protein levels and one of the 5 had OCB. Multifocal white matter lesions were seen in 4, with contrast enhancement in 2, but one showed diffuse white matter lesions (figure 4E), which is atypical of MS. Spinal cord lesions were present in 2 of 5 on MRI. Visual evoked potentials were abnormal in 4 of 5 patients examined. Four met the McDonald criteria, while one did not. All fulfilled the established criteria for CIDP (figure 4, F and G). In the 5 anti-neurofascin antibody–positive CCPD patients, corticosteroids were only partially effective or ineffective except in one with a favorable response of the CNS lesions. IVIg administrations were effective in all 4 patients that received them, while plasma exchanges were effective in 2 of 3. These clinical features were similar to those of the 2 anti-neurofascin antibody–negative CCPD patients; however, in both anti-neurofascin antibody–seronegative CCPD patients, corticosteroids were effective for both CNS and PNS lesions.

DISCUSSION This study reveals the high frequency of anti-neurofascin antibody in patients with CCPD. We also confirmed that some patients with MS, CIDP, AMAN, and AIDP were positive for anti-neurofascin antibodies by ELISA, as reported previously. The discrepancies among the results of the ELISA, immunoblotting, and cell-based assays may be attributable to differences in epitope, i.e., linear or conformational, as well as the sensitivities and specificities of the 3



(A–D) MRIs for patient 1. (A, B) Brain MRI fluid-attenuated inversion recovery images showing MS-like lesions in juxtaven-tricular regions. (C) The cauda equina is contrast-enhanced (gadolinium-enhanced T1-weighted image). (D) Enlarged cauda equina (T2-weighted image). (E) Brain MRI (fluid-attenuated inversion recovery image) for patient 2. Diffuse white matter lesions are seen. (F, G) Nerve conduction study findings for patients 1 (F) and 2 (G). Severe conduction blocks with reduced amplitudes and temporal dispersions are evident in these patients with CCPD. CCPD = combined central and peripheral demyelination; MS = multiple sclerosis.

assays. Further refinement of the assay methods and a large-scale study are called for to solve this issue in the future. As CCPD is an extremely rare disease, only 3 of the patients described were our own institutional cases. Although selection bias was not completely eliminated, on referral to our clinic for anti-neurofascin antibody assay, no selection was made; therefore, we believe that the recruitment of patients with CCPD would not have severely distorted our findings.

With these reservations in mind, the following are characteristic of CCPD patients with anti-neurofascin antibodies: 1) CNS and PNS involvement occur either simultaneously or sequentially with a short or long interval; 2) peripheral nerve demyelination is indistinguishable from CIDP; 3) CNS involvement is mostly typical for MS, in which spinal cord lesions and gadolinium enhancement of the lesions can develop, but occasionally atypical, demonstrating diffuse cerebral white matter lesions; 4) CSF OCB are negative in most cases while CSF protein levels show variable degrees of increase; and 5) the response to corticosteroids is limited, while IVIg and plasma exchanges are beneficial for both CNS and PNS lesions.

In a previous report, spreading of autoimmunity from central to peripheral or peripheral to central myelin was proposed in patients with either MS or CIDP.⁸ However, 3 of 5 patients with CCPD who were positive for anti-neurofascin antibodies by the

cell-based assay showed simultaneous or sequential occurrence of both CNS and PNS lesions within a short interval, suggesting that the anti-neurofascin antibody can emerge at the onset or very early in the course of the disease, which is contrary to the epitope spreading hypothesis. By contrast, in the other 2 anti-neurofascin antibody-positive cases with a long interval between CNS and PNS lesion development, anti-neurofascin antibody could be produced by epitope spreading. Nonetheless, such secondarily produced antibody may exert pathogenic actions extending the original lesions from the PNS to the CNS or vice versa. The presence of anti-neurofascin antibody in the CSF further supports its involvement in the formation of CNS lesions in addition to the PNS lesions. Anti-neurofascin antibody-positive CCPD patients fulfilled the European Federation of Neurological Societies/Peripheral Nerve Society diagnostic criteria for CIDP.¹⁰ By contrast, CNS involvement was atypical in one anti-neurofascin antibody-positive patient who had a chronic course and showed diffuse white matter changes on brain MRI. Of interest is that many reported patients with CCPD, including our antibody-positive patients, were negative for OCB, which are detected in >90% of patients with MS.¹¹ These findings suggest that anti-neurofascin antibody-positive CCPD patients could have a somewhat distinct CNS

involvement from that in typical MS patients. It is thus suggested that anti-neurofascin antibody–positive CCPD is a unique condition due to a common immunopathogenic mechanism between CNS and PNS demyelination rather than a simple coincidence of MS and CIDP.

An autoantibody response to neurofascin has been reported in patients with MS, particularly in those with chronic progressive MS, and immune-mediated axonal injury and a potential contribution to axonal pathology have been proposed.¹² Autoantibody responses to neurofascin were also reported in patients with GBS and CIDP.^{13–15} Among the patients with GBS and CIDP in that study, only 4% were positive for neurofascin by a cell-based assay, although CNS involvement was not thoroughly examined in the positive cases.¹⁴ Two patients with CIDP who had high titers of anti-neurofascin antibody by ELISA benefited from plasma exchange.¹⁴ Our findings are partly in accordance with the observations described in a recent report,¹⁴ and thus suggest that a high titer of anti-neurofascin antibodies recognizing neurofascin molecules expressed on the cell membrane could also be associated with CCPD, and might be pathogenic. In this case, the CNS and PNS are possibly involved together, because neurofascin exists in both tissues. In a small fraction of AMAN, AIDP, CIDP, and MS cases with high titers of anti-neurofascin antibody, as reported previously,^{12–15} investigations for subclinical involvement of the CNS and PNS may be recommended.

Neurofascins are transmembrane adhesion molecules expressed at the nodes and paranodes of both the CNS and PNS.^{5,16,17} The 2 major isoforms of neurofascin have different expression patterns. NF186 is an axonal membrane protein contributing to assembly of voltage-dependent sodium channels at high concentrations at the node of Ranvier. NF155 is expressed by glial cells (oligodendrocytes and Schwann cells) at the paranodes and connects the myelin sheath to the axon via binding to contactin-1 and Caspr. The autoantibodies in patients with CCPD appear to recognize mainly NF155 that is present at the paranode based on the findings of the cell-based assay and immunoabsorption experiments; however, some reactivity against the node was also observed in immunohistochemistry experiments using PNS tissue sections, suggesting additional binding to NF186 at the nodal axolemma in situ, at least in some patients. Our findings indicate that autoantibodies attacking axo-glial integrity at the paranodes and nodes, but not compacted myelin, may cause inflammatory demyelination in both the CNS and PNS. The beneficial responses to IVIg and plasma exchanges in patients with CCPD who are anti-neurofascin antibody positive also support the notion that this autoantibody may contribute to the demyelinating process. Thus, the results of our study

suggest that the presence of anti-neurofascin antibodies is a potential diagnostic biomarker for patients with CCPD who could benefit from IVIg and plasma exchanges.

AUTHOR CONTRIBUTIONS

Dr. Kawamura and Dr. Yamasaki drafted the manuscript for content, collected and statistically analyzed data. Dr. Yonekawa, Dr. Matsushita, Dr. Kusunoki, Dr. Nagayama, Dr. Fukuda, Dr. Ogata, and Dr. Matsuse collected data. Dr. Murai revised the manuscript for content. Dr. Kira revised the manuscript for content, obtained the funding for this study, and supervised the study.

STUDY FUNDING

Supported in part by a Health and Labour Sciences Research grant on intractable diseases (H24-Nanchitou [Nan]-Ippan-055 and H23-Nanchi-Ippan-017) from the Ministry of Health, Labour and Welfare, Japan; by a Scientific Research B grant (no. 22390178); and a Challenging Exploratory Research grant (no. 23659459) from the Ministry of Education, Culture, Sports, Science and Technology, Japan.

DISCLOSURE

N. Kawamura, R. Yamasaki, and T. Yonekawa report no disclosures. T. Matsushita has received speaker honoraria from Bayer Schering Pharma, Biogen Idec, and Pfizer and receives research support from Bayer Schering Pharma, the Ministry of Health, Labour and Welfare of Japan, the Japan Science and Technology Agency, the Ministry of Education, Culture, Sports, Science and Technology of Japan, and the Kaibara Morikazu Medical Science Promotion Foundation, Japan. S. Kusunoki serves as an editorial board member of *Experimental Neurology*, *Journal of Neuroimmunology*, and *Clinical & Experimental Neuroimmunology*. He received honoraria from Teijin Pharma Limited, Nihon Pharmaceuticals Co. Ltd., and Benesis Corporation. He is funded by research grants from the Ministry of Health, Labour and Welfare, Japan, and grants from the Ministry of Education, Culture, Sports, Science and Technology, Japan. S. Nagayama, Y. Fukuda, H. Ogata, and D. Matsuse report no disclosures. H. Murai serves as associate editor of *BMC Neurology*, and receives research support from the Ministry of Health, Labour and Welfare, Japan (H22-nanchi-shitei-002). J. Kira serves as an editorial board member of *Multiple Sclerosis Journal*, *Multiple Sclerosis and Related Disorders*, *BMC Medicine*, *PLoS ONE*, *Expert Review of Neurotherapeutics*, *Intractable and Rare Diseases Research*, *The Scientific World Journal*, and *Journal of the Neurological Sciences*. He is a consultant for Biogen Idec Japan, and has received honoraria from Bayer Healthcare and funding for a trip from Bayer Healthcare and Biogen Idec Japan. He is funded by a research grant for nervous and mental disorders from the Ministry of Health, Labour and Welfare, Japan, and grants from the Japan Science and Technology Agency and the Ministry of Education, Culture, Sports, Science and Technology, Japan. Go to Neurology.org for full disclosures.

Received December 30, 2012. Accepted in final form May 6, 2013.

REFERENCES

1. Kamm C, Zettl UK. Autoimmune disorders affecting both the central and peripheral nervous system. *Autoimmun Rev* 2012;11:196–202.
2. Zephir H, Stojkovic T, Latour P, et al. Relapsing demyelinating disease affecting both the central and peripheral nervous systems. *J Neurol Neurosurg Psychiatry* 2008;79:1032–1039.
3. Pirko I, Kuntz NL, Patterson M, Keegan BM, Weinshenker BG, Rodriguez M. Contrasting effects of IFNbeta and IVIG in children with central and peripheral demyelination. *Neurology* 2003;60:1697–1699.
4. Thomas PK, Walker RW, Rudge P, et al. Chronic demyelinating peripheral neuropathy associated with multifocal central nervous system demyelination. *Brain* 1987;110(pt 1):53–76.

5. Kriebel M, Wuchter J, Trinks S, Volkmer H. Neurofascin: a switch between neuronal plasticity and stability. *Int J Biochem Cell Biol* 2012;44:694–697.
6. Research criteria for diagnosis of chronic inflammatory demyelinating polyneuropathy (CIDP). Report from an Ad Hoc Subcommittee of the American Academy of Neurology AIDS Task Force. *Neurology* 1991;41:617–618.
7. Polman CH, Reingold SC, Banwell B, et al. Diagnostic criteria for multiple sclerosis: 2010 revisions to the McDonald criteria. *Ann Neurol* 2011;69:292–302.
8. Ho TW, Mishu B, Li CY, et al. Guillain-Barré syndrome in northern China: relationship to *Campylobacter jejuni* infection and anti-glycolipid antibodies. *Brain* 1995;118(pt 3):597–605.
9. Pomfret AD, Shroff SM, Fuss B, et al. Novel forms of neurofascin 155 in the central nervous system: alterations in paranodal disruption models and multiple sclerosis. *Brain* 2010;133:389–405.
10. European Federation of Neurological Societies/Peripheral Nerve Society Guideline on management of chronic inflammatory demyelinating polyradiculoneuropathy. Report of a joint task force of the European Federation of Neurological Societies and the Peripheral Nerve Society. *J Peripher Nerv Syst* 2005;10:220–228.
11. Link H, Huang YM. Oligoclonal bands in multiple sclerosis cerebrospinal fluid: an update on methodology and clinical usefulness. *J Neuroimmunol* 2006;180:17–28.
12. Mathey EK, Derfuss T, Storch MK, et al. Neurofascin as a novel target for autoantibody-mediated axonal injury. *J Exp Med* 2007;204:2363–2372.
13. Lonigro A, Devaux JJ. Disruption of neurofascin and gliomedin at nodes of Ranvier precedes demyelination in experimental allergic neuritis. *Brain* 2009;132:260–273.
14. Ng JK, Malotka J, Kawakami N, et al. Neurofascin as a target for autoantibodies in peripheral neuropathies. *Neurology* 2012;79:2241–2248.
15. Devaux JJ, Odaka M, Yuki N. Nodal proteins are target antigens in Guillain-Barré syndrome. *J Peripher Nerv Syst* 2012;17:62–71.
16. Ratcliffe CF, Westenbroek RE, Curtis R, Catterall WA. Sodium channel beta1 and beta3 subunits associate with neurofascin through their extracellular immunoglobulin-like domain. *J Cell Biol* 2001;154:427–434.
17. Sherman DL, Tait S, Melrose S, et al. Neurofascins are required to establish axonal domains for saltatory conduction. *Neuron* 2005;48:737–742.
18. Tait S, Gunn-Moore F, Collinson JM, et al. An oligodendrocyte cell adhesion molecule at the site of assembly of the paranodal axo-glia junction. *J Cell Biol* 2000;150:657–666.

Do You Know What is Happening to Neurology on Capitol Hill?

Congress is making decisions that affect neurologic research funding and the way neurology is practiced in the United States. Only *Capitol Hill Report* on AAN.com takes you behind Washington's closed doors and shines a light on how your federal legislators are working for—or against—your interests. Read *Capitol Hill Report* on AAN.com the second and fourth Monday of each month. Stay informed. Your work depends on it.

We're Looking for Our Next Generation of Leaders

Applications are now being accepted to the American Academy of Neurology's 2013 Emerging Leaders Forum. The program is designed to identify, orient, and cultivate future leaders of the world's largest association of neurologists. Applicants must be members of the AAN 45 years of age or younger—or no more than 13 years out of training.

Apply by September 1, 2013.

www.aan.com/view/leadership or contact Donna Honeyman at (612) 928-6055 or dhoneyman@aan.com.